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# Problem J. Subrectangle Count

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|---|---|
| Input file: | standard input |
| Output file: | standard output |
| Time limit: | 2 seconds |
| Memory limit: | 1024 megabytes |

Given an integer sequence of length $n(n \geq 2)$, denoted as $a_{1,2,\ldots,n}$, and an integer sequence of length $m(m \geq 2)$, denoted as $b_{1,2,\ldots,m}$, we can construct an $n \times m$ matrix $c$ such that $c_{i,j} = a_i \oplus b_j (1 \leq i \leq n, 1 \leq j \leq m)$, where $\oplus$ denotes the bitwise XOR operation.

Determine how many pairs $(i,j)(1 \leq i < n, 1 \leq j < m)$ satisfy that the subrectangle region of $c$ given by $\begin{bmatrix} c_{i,j} & c_{i,j+1} \\ c_{i+1,j} & c_{i+1,j+1} \end{bmatrix}$ is of the form $\begin{bmatrix} x & x+1 \\ x+2 & x+3 \end{bmatrix}$, meaning that $c_{i,j+1} = c_{i,j} + 1$, $c_{i+1,j} = c_{i,j} + 2$, and $c_{i+1,j+1} = c_{i,j} + 3$.

## Input

The first line contains an integer $T(1 \leq T \leq 10^5)$, representing the number of test cases.

For each test case:

The first line contains two space-separated integers $n, m(2 \leq n, m \leq 2 \times 10^5)$, representing the number of rows and columns of $c$.

The second line contains $n$ space-separated integers, representing $a_1, a_2, \ldots, a_n (0 \leq a_i < 2^{30})$.

The third line contains $m$ space-separated integers, representing $b_1, b_2, \ldots, b_m (0 \leq b_i < 2^{30})$.

It is guaranteed that the sum of $n$ does not exceed $2 \times 10^5$ and the sum of $m$ does not exceed $2 \times 10^5$.

## Output

For each test case, output a single line containing an integer, representing how many pairs $(i,j)$ satisfy the condition.

## Example

| standard input | standard output |
|---|---|
| 5<br>2 2<br>0 2<br>0 1<br>3 3<br>6 4 2<br>6 7 10<br>5 4<br>4 6 5 1 3<br>1 0 2 3<br>7 2<br>7 3 1 4 0 0 1<br>6 5<br>5 5<br>1 7 5 3 1<br>3 2 7 7 6 | 1<br>2<br>1<br>0<br>6 |